

Name :

Class :

StudentID :

Part A

Please draw both the E-R (Entity-Relationship) Diagram. The descriptions below are sourced from the textbook.

University Database Requirements

A university plans to develop a database to manage its colleges, departments, instructors, students, courses, and course sections, and to record student enrollments and grades. The requirements are as follows:

Each college (COLLEGE) has a unique name (CName). A college may administer zero or more departments, and each department must belong to exactly one college.

Each department (DEPT) has a unique department name (DName) and a unique department code (DCode).

Each instructor (INSTRUCTOR) has a unique identifier (Id), a name (IName), and a phone number (IPhone).

Each instructor must work for exactly one department. A department may employ zero or more instructors.

Each department must have exactly one instructor serving as its chair. An instructor may chair zero or one department. The database also records the date on which the instructor began serving as chair (CStartDate).

Each student (STUDENT) has a unique student ID (SId) and a name (SName). The name consists of a first name (FName) and a last name (LName).

A student may belong to zero or one department. A department may have zero or more students.

Each course (COURSE) has a unique course code (CCode) and a unique course name (CoName).

Each course must be offered by exactly one department. A department may offer zero or more courses.

A course may have zero or more sections (SECTION), and each section must belong to exactly one course.

Each section must be taught by exactly one instructor. An instructor may teach zero or more sections.

A student may enroll in zero or more sections. Each section must have at least five students enrolled.

The database records each student's grade (Grade) for a section as an attribute of the enrollment relationship (TAKES) between STUDENT and SECTION.

E-R diagram

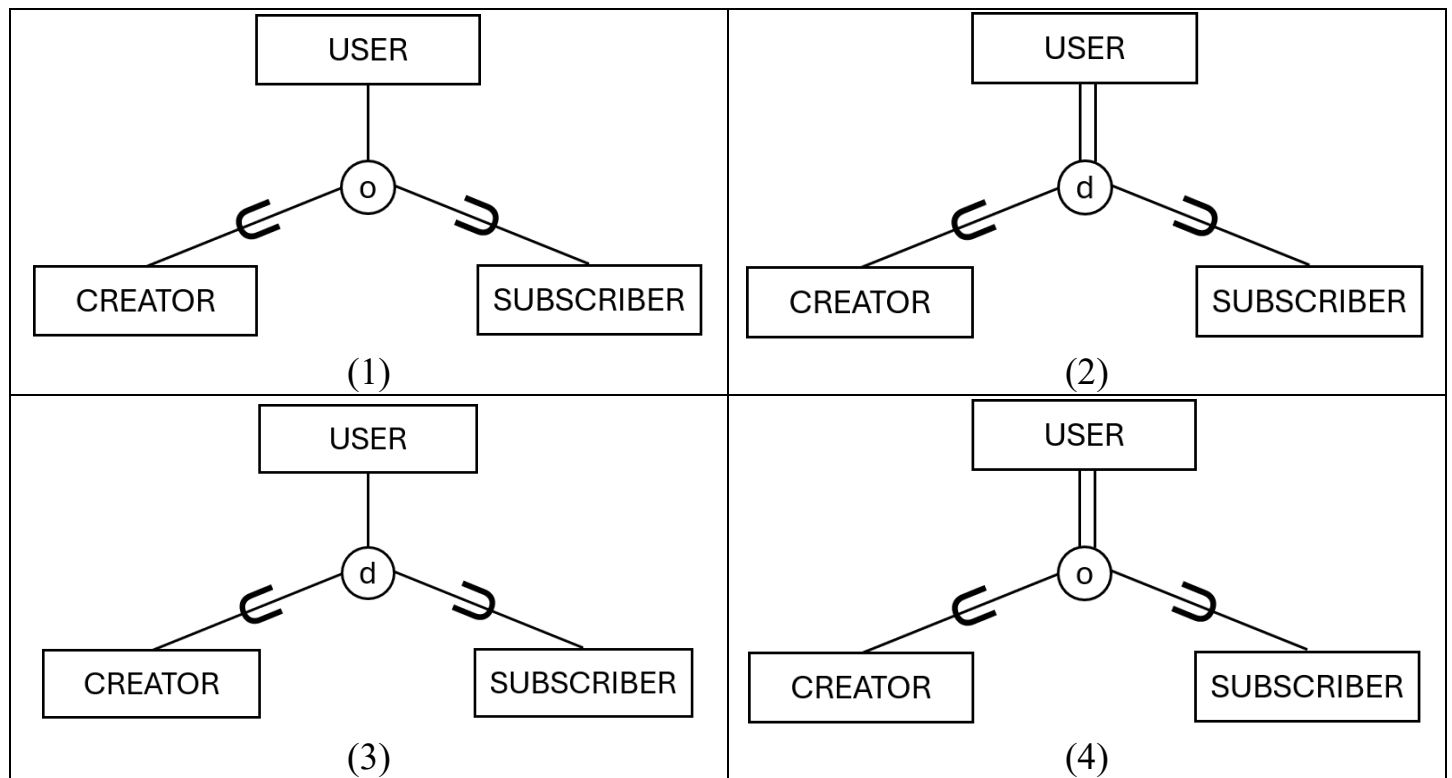
Please answer on the answer sheet

Part B

ER Diagram Matching Exercise

Match each of the four diagrams with the statements labeled A–F. Select **one statement for each diagram**. Each statement may be used only once, and **two statements will remain unused**.

- A. A user may be a creator, a subscriber, or neither, but cannot be both.
- B. Every user must be either a creator or a subscriber, but cannot be both.
- C. Anyone who registers on the platform must both create content and subscribe to others.
- D. Every user must be a creator, a subscriber, or both; no user can be neither.
- E. The **o** in the diagram means that users who create content cannot also subscribe to others.
- F. A user may both create content and subscribe to others, or simply register and browse without being either a creator or a subscriber.



Matching Exercise

Please answer on the answer sheet, clearly indicating the question number for each answer.